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THE FORMATION OF TRANSNATIONAL KNOWLEDGE NETWORKS ON SOCIAL MEDIA

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ABSTRACT: *Transnational knowledge networks provide organizations with information useful in addressing shared problems. Social media may enable the formation of those networks, yet their role in the process has received little attention. This article examines the structure and antecedents of two networks facilitated by the microblogging platform Twitter operating in the policy domain of emergency management. One network includes national-level government agencies responsible for disaster response and recovery operations; the other includes nongovernmental organizations in the form of Red Cross and Red Crescent national societies. We use a logistic regression quadratic assignment procedure to test hypotheses derived from related literature. While findings indicate that shared language and geographic proximity shaped network formation, both networks exhibit boundary spanning behavior in which organizations sought out information from high-profile, resource-rich organizations. Those organizations helped to connect otherwise regionally bound clusters and demonstrate the nascent potential of social media to create global transnational knowledge networks.*

INTRODUCTION

Information sharing is an important type of interorganizational cooperation because it offers decision makers guidance and evidence upon which to base decisions and improve the efficiency and efficacy of operations (Dawes, Cresswell, and Pardo 2009; Fountain 2001; Gil-Garcia 2012; Reddick 2008). Such interactions across national boundaries create the basis for transnational knowledge networks in which multiple actors participate and draw from shared sources to address common problems (Dawes, Gharawi, and Burke 2012). These networks exist in a variety of policy domains, such as border security, immigration, and trade (Navarrete et al. 2010); environmental protection (Cresswell, Burke, and Navarrete 2009); and disaster response and recovery (Comfort, Boin, and Demchak 2010).

Dawes et al. (2012, 114) defined transnational knowledge networks as two or more organizations “located in different countries involved in the exchange of knowledge, information, or both in order to address a mutual concern,” and attributed their formation to a variety of factors, including common organizational goals, culture, shared language, and geographic proximity. This type of network formation model, however, has not been empirically tested beyond small-N case studies bound by geographic region (Gil-Garcia 2012; Cresswell et al. 2009). Thus, our understanding of global trends is limited.

This article examines the extent to which knowledge networks emerge globally through social media and tests key factors. We investigate worldwide network formation across two sets of organizations operating in the policy domain of emergency management: (a) government agencies tasked as their country’s lead for disaster response and recovery operations; and (b) Red Cross and Red Crescent national societies. Other research explores knowledge networks as either government-based (Dawes et al. 2012; Gil-Garcia, Chengalur-Smith, and Duchessi 2007) or NGO-based (Dumoulin 2003). We maintain this distinction in part to account for the impact of a strong proponent of social media in one network [e.g., the International Federation of Red Cross and Red Crescent Societies (IFRC)] versus the lack of one in the government network. The policy domain was chosen due to the fact that shared risk creates a demand for shared information (Comfort 1999), and emergency management organizations operate across a range of operational and administrative demands that often require collaboration (Boin and Rhinard 2008).

In both cases, we find that while shared language and geographic proximity shaped and constrained network formation, evidence points to boundary spanning behavior in which organizations sought out information from high-profile, resource-rich organizations regardless of language and geographic location. Although these central organizations represent hubs that link otherwise disparate actors, both networks demonstrate limited participation by most organizations. We conclude by discussing those limitations and how social media platforms such as Twitter allow organizations to sidestep the acquisition of proprietary information systems and overcome barriers to information sharing.

Transnational Knowledge Networks and Social Media

From local agencies to transnational nongovernmental organizations (TNGOs), personnel increasingly recognize the importance of knowledge sharing and collaboration (Mitchell, O’Leary, and Gerard 2015). As in other types of networks, managers work “to foster the transfer, receipt, and integration of knowledge across the network and, ultimately, to build a long-term collaborative problem-solving capacity” (Weber and Khademian 2008, 334). Organizations then use this knowledge and information to address public problems that no one organization can single-handedly manage (Dawes et al. 2009; Gil-Garcia 2012).

The adoption of information technology (IT) supports that type of capacity building (Fountain 2001; Mergel 2012; Roberts 2011).¹ Transnational knowledge networks represent the manifestation of sociotechnical systems and are facilitated by a combination of (a) trusted social networks; (b) shared information; (c) integrated data; and (d) an interoperable technical infrastructure (Gil-Garcia, Pardo, and Burke 2010, 197). Such systems are generally Web-based and closed-access, which requires participants to negotiate terms of participation (Chatfield and AlAnazi 2015; Gottschalk and Solli-Saether 2009), which thus further complicates the network formation process (Fishenden and Thompson 2013).

Social media platforms such as the microblogging site Twitter represent networking platforms through which users generate and share content (Mergel 2012). Their easy accessibility circumvents traditional barriers associated with proprietary systems (Reddick and Norris 2013). The microblogging website Twitter is particularly popular. Government agencies and nongovernmental organizations around the world have established Twitter accounts (Bonsón, Royo, and Ratkai 2015; Nah and Saxton 2013; Wukich and Khemka 2016), and by doing so have gained the ability to engage with the public and other agencies via those channels (Meijer 2012; 2014). Twitter conventions—such as following a user’s content, mentioning a user, and reposting (i.e., retweeting) content from others—facilitate basic dyadic relationships that aggregate to form larger information or knowledge networks (Southwell 2013; Sutton et al. 2013).

While a growing body of research addresses how government actors engage the public via social media (Bryer and Nelson 2013; Meijer, Grimmelikhuijsen, and Brandsma 2012; Mossberger, Wu, and Crawford 2013; Noveck 2009), less is known about how government and NGOs collectively engage each other, particularly the creation and maintenance of transnational knowledge networks. Mergel (2012) noted how wikis (i.e., collaborative websites) were used to facilitate cooperation in the US in a number of cases at the federal level. Sutton et al. (2013) illustrated an interagency information network that emerged on Twitter following the 2010 BP Oil Spill off the coast of Louisiana (USA), but did not address the factors that influenced its formation. We expand on this research by drawing on findings from e-government, collaborative governance, transnational nongovernmental organizations, and international political economy to generate a set of hypotheses regarding what factors drive network formation in this context. These disparate fields offer

insights on possible factors that influence both informal and formal knowledge sharing arrangements across national borders. By testing those hypotheses, we contribute to the understanding of how knowledge networks form globally and how social media can facilitate that process. We do so by examining networks created by organizations following the content of other organizations on Twitter.

Factors Influencing Knowledge Network Formation

Knowledge networks address common problems, often within single policy domains (Gil-Garcia 2012); therefore, the recognition of shared problems is critical. For example, Creswell et al. (2009) illustrated examples related to border security in which the United States created partnerships with Mexico and Canada to identify high-risk travelers, exchange criminal records, and adopt compatible biometric technologies. Emergency management is another common concern that organizations address through information sharing agreements and IT systems (Comfort et al. 2010; Kapucu 2006; Mendonça 2007). It represents a policy domain in which shared risk creates a demand for shared information and collaboration (Boin and Rhinard 2008; Comfort 1999).

During disasters, organizations seek out information and resources to reduce their constituents' exposure to risk, alleviate human suffering, and maintain continuity of operations (Comfort et al. 2010). In these networks, many turn to resource-rich organizations that possess the capacity to provide assistance and high-quality information (Comfort 1999; Kapucu 2006; 2009). In these uncertain environments, participants rely on the tested public information procedures used to generate and distribute trusted guidance, advice, and situational awareness (Hughes and Palen 2012). This tendency has been formalized through emergency support function arrangements in the United States and United Nations clusters abroad in which lead agencies coordinate activities based on the availability of resources and task competency (Jahre and Jensen 2010; Waugh 2009).

In other policy domains, the degree to which actors consider information to be accurate, reliable, and of high quality also represents an important factor in whether they engage in knowledge network activities (Gil-Garcia et al. 2010). Wilensky (1967) identified the characteristics of high-quality information; in addition to accuracy and reliability, he noted that information should be timely, clear, and comprehensive.

Wealthier countries may possess more resources to develop and disseminate high-quality information. Countries with higher per capita GDP, for example, have relatively greater capacities to protect populations by investing in effective mitigation strategies (Kahn 2005) and robust response and recovery infrastructures (Kapucu and Liou 2014). That capacity also facilitates the development of an effective online presence. Wukich and Khemka (2016), for example, observed that wealthier countries were more likely to adopt social media for emergency management purposes and were more apt to develop high quality messages. Therefore, we expect to observe lower-income countries following higher-income countries in search of best practices and other pieces of quality information in line with the logic of

resource dependence as an antecedent to interorganizational cooperation (Gulati and Garguilio 1999; Pfeffer and Salancik 1978). While following a Twitter account requires very limited effort and resources, developing content that might interest others on a consistent basis does require considerable time, effort, and resources (Mergel 2012); therefore, we would assume that others would want to follow users who disseminate valuable content.

In addition to country-level variables, social-media-related attributes may offer indications of organizational capacity and quality. On Twitter, for example, follower counts may signal active agencies that provide high-quality information. Furthermore, organizations using several different types of social media may be seen as more active and professional, signaling higher levels of capacity.

H1: Organizations are more likely to follow Twitter message content of organizations with higher levels of capacity than organizations with lower levels of capacity.

Dawes et al. (2012) claimed that network formation and performance were influenced by the ability of organizations to reduce the contextual distances—related to mission and culture—between themselves and potential partners. We suspect that agencies with similar operational characteristics, therefore, will be more likely to share information. Organizational mission is one such attribute. Agencies focusing solely on emergency management might be more likely to follow one another because their interests converge on a single policy domain, in contrast to a combination of emergency management and other areas such as domestic security, public health, public safety, and/or environmental protection. They could then share best practices and instruction without filtering the noise of unwanted messages.

H2: Organizations focusing solely on emergency management are more likely to follow other organizations that focus solely on emergency management rather than organizations that address additional policy domains.

Conversely, proponents of IT speculate that adoption and use will lead to patterns of information exchange that transcend traditional centralized structures and create more diverse networks (Mergel 2012; Roberts 2011; Fountain 2001). The boundaries of what constitutes a diverse network in emergency management may include domestic security, public health, public safety, and/or environmental protection (Comfort et al. 2010), and social media may offer organizations with unprecedented access to such information. Wukich and Mergel (2016), for example, reported that emergency management agencies in the United States frequently retweeted message content from organizations from other policy domains (e.g., the earth sciences, law enforcement, and public health); therefore, it is possible to observe network relationships not bound by strict policy domain affiliations.

Navarrete et al. (2010) observed that common language facilitates network formation in international settings. Language barriers, conversely, can discourage or interrupt collaboration; for example, Roth and Sever (2007) noted this problem in the context of domestic security and anti-terrorism campaigns. It may follow then that agencies will avoid others that do not share the same primary language.

H3: Organizations are more likely to follow organizations operating in the same language rather than organizations operating in a different language.

It also stands to reason that the English language may be more popular than others. English is the most spoken second language worldwide; therefore, patterns of English-language followership may be prominent.

As in other forms of cooperation, physical location plays a role. Geographic proximity has been shown positively to affect trade between states, for example (Egger 2002; Leamer and Levinsohn 1995); the closer the two states, the lower their transportation costs and the more likely they are to engage in commerce (Gowa 1994). With respect to information sharing, Sonn and Storper (2008) demonstrate the continued importance of geographical proximity between firms for knowledge production and innovation purposes. At the sub-state level, closer geographical proximity has been shown to promote interlocal cooperation (Lee, Feiock, and Lee 2012), connections between TNGOs (Shumate, Fulk, and Monge 2005), and scientific collaboration across social sectors (Katz 1994). The theoretical intuition here is that organizational knowledge tends to accumulate locally due to cost concerns (Jaffe, Trajtenberg, and Henderson 1993; Rosenkopf and Almeida 2003).

Networks facilitated via IT present interesting challenges to those theories. The cost concerns of one organization's attempt to gain global knowledge are reduced considerably (Fountain 2001; Mergel 2012). Despite the flattening effects of IT, geographic proximity between two organizations still may influence the decision to engage in knowledge sharing networks because formal agreements may require in-person negotiations (Cummings and Teng 2003). Dawes et al. (2012) explained, "When participants are widely dispersed across geographic areas, effective communication among people in different locations and smooth transmission of information and knowledge may be hindered" (117). We assume, therefore, that organizations within closer geographic proximity are still more likely to engage in knowledge exchange because they share more immediate public problems such as the exposure to shared risk (Comfort 1999).

H4: Organizations are more likely to follow organizations that are geographically closer to them rather than organizations that are geographically farther away.

Not all knowledge sharing agreements originate from the actual participants. External organizations at times play steering and/or boundary spanning roles

(Boardman 2011; Kapucu 2006). For example, Jandhyala and Phene (2015) described intergovernmental organizations as a means to “generate commonalities among participants and promote a set of similar rules, expectations, and norms across different countries and thus may be effective in supporting cross-border knowledge transfer” (712). In addition, Lance, Georgiadou, and Bregt (2009) highlighted the role of federal budget agencies in the US and Canada in promoting geospatial information sharing.

Adopting social media and engaging in transnational knowledge networks represents a change in operations for many organizations. At the federal level in the US, Kelman (2005) found that the presence of a high-profile champion increased the likelihood of adopting change. Mergel and Bretschneider (2013) illustrate a process in which organizations adopt social media where individuals first informally experiment with the technology and apply lessons learned from their external environment. We expect, therefore, that the presence of external agencies, such as the IFRC, advocating for social media adoption will increase the likelihood of participation in knowledge sharing networks, whereas organizations not exposed to social media advocates will be less likely to participate.

H5: Organizations are more likely to follow others and engage in knowledge sharing in networks in which an external champion for social media adoption and use is active.

These hypotheses enable us to examine the question of what factors influence transnational network formation via a social media.

METHODOLOGY

Case Selection

To test our hypotheses, we chose to examine country-level organizations operating around the world in the policy domain of emergency management and the extent to which they follow each other's content on Twitter, creating transnational knowledge networks. The follower relationship represents an appropriate measure for knowledge sharing because government agencies routinely monitor social media to gauge the public's mood (Zavattaro, French, and Mohanty 2015), interact with constituents (Meijer 2014), and gain insights from other organizations (Sutton et al. 2013). NGOs demonstrate similar behavior (Brionnes et al. 2011). Specifically germane to emergency management, social media monitoring helps organizations to identify risks and maintain situational awareness at all times (St. Denis, Palen, and Anderson 2014; Sutton 2009). Furthermore, a growing number of organizations in the US (Su, Wardell, and Thorkildsen 2013), Europe (Reuter et al. 2016), and in developing economies such as Indonesia (Meier 2015) regularly monitor their social media feeds to accrue situational awareness. Therefore, the act of following an organization creates a direct communication channel between organizations.

We analyze two networks: (a) national-level government agencies tasked with emergency management responsibilities; and (b) Red Cross and Red Crescent national societies—the organizations that provide additional support to those countries. Consequently, we were able to model both public sector and nongovernmental knowledge networks. The public sector network includes disparate government agencies with a range of available resources and organizational missions—some agencies implement only emergency management policy, while others also conduct work related to domestic security, public health, public safety, and/or environmental protection. Red Cross and Red Crescent national societies are loosely linked through their association with the IFRC, and the IFRC has provided organizations with social media best-practices and training opportunities (IFRC 2009). The government agencies, conversely, did not have a comparable, high-profile external advocate for social media adoption and use.

Data Collection

Our dependent variable is whether an organization seeks out information from another via social media, and we operationalize it as whether an emergency management organization follows the Twitter account of another emergency management organization in a foreign country. Thus, our unit of analysis is the directional dyad. We collect data for this variable according to the following process: First, we identified government emergency management organizations via the United Nations Office for Disaster Risk Reduction and other organizational websites.² We also identified Red Cross and Red Crescent national societies from the International Federation of Red Cross and Red Crescent (IFRC) website—185 government-level emergency management organizations, and 193 Red Cross and Red Crescent emergency management organizations. Second, we reviewed each organization's official website for advertised Twitter accounts. Although other types of social media were also used by the organizations that we studied (e.g., Vkontakte in Russia and Weibo in China), Twitter has been adopted across all geographic regions, and unlike Facebook, it makes available data to conduct network analysis. Reviewing official websites reduced the risk of selecting unofficial or false accounts (Sutton et al. 2013; Wukich and Mergel 2015). For accounts not advertised on organizations' official sites, we conducted searches via Google and Twitter in both English and the home country's language and script. Third, with these Twitter accounts, we used the software program NodeXL (Smith et al. 2010) to determine, for each organization, whether or not it followed other organizations identified in the public sector or NGO networks. Those relationships provide the basis for the two larger knowledge networks.

Table 1 reports the frequency distribution of Twitter adoption by geographic region. In total, 60 government agencies (32.4% of all relevant agencies) used Twitter, with the highest rates in the Americas (40.4%), Asia (38.3%), and Europe (31.9%). Conversely, 50.3% of all Red Cross and Red Crescent national societies (97) had adopted the social media platform, with particularly high rates in the Americas (62.2%) and Asia (61.3%). While the digital divide is particularly evident

TABLE 1
Twitter Adoption Rates by Geographic Region

	<i>Government Agencies</i>			<i>Red Cross/Crescent</i>		
	<i>Orgs</i>	<i>Twitter</i>	<i>%</i>	<i>Orgs</i>	<i>Twitter</i>	<i>%</i>
Africa	47	6	12.8	53	18	34.0
Americas	38	19	40.4	37	23	62.2
Asia	46	18	38.3	49	30	61.3
Europe	42	15	31.9	43	23	53.5
Oceania	12	2	4.3	11	3	27.3
Total	185	60	32.4	193	97	50.3

in Africa and the Oceania region, a notable number of organizations have adopted Twitter across the world. The IFRC’s social media advocacy efforts may help to explain why the Red Cross and Red Crescent adoption rates are higher than that of the government agencies. Relatedly, unlike government agencies, NGOs may be more inclined to use social media for fundraising and volunteer recruitment purposes, which may increase usage (Briones et al. 2011).

We also collected individual and dyadic attribute data for the organizations involved in the networks to operationalize our explanatory and control variables from the following sources: the United Nations (e.g., geographic regions and sub-regions), the CIA World Factbook (e.g., GDP per capita), Twitter (e.g., follower counts), Internet World Stats (e.g., percent of population with Internet access), OraGIS (e.g., the geographic distance between each organization), and organizational websites (e.g., primary language; whether organizations pursued solely an emergency management mission; and a social media index calculated by adding the number of accounts advertised on an organization’s website, including Twitter, Facebook, YouTube, LinkedIn, Flickr, Google+, and other platforms such as Vkontakte and Weibo). We directly link these variables to our hypotheses in the data analysis section. Of note, organizational mission captures the variability in task functions that organizations execute. Agencies focusing solely on emergency management might be more likely to follow one another because their interests converge on a single policy domain, in contrast to a combination of emergency management and other areas such as domestic security, public health, public safety, and/or environmental protection.

Table 2 provides a summary of the independent variables included in our model and their respective hypotheses.

Data Analysis

Our ultimate goal was to identify the factors that facilitated or constrained the likelihood that one organization followed another organization via Twitter. Because these organizations operate within networks, the assumption that each tie is

TABLE 2
Hypotheses and Independent Variables

<i>Hypothesis</i>	<i>IV</i>	<i>Data Type</i>	<i>Expected Effect</i>
<i>Homophily Effects</i>			
H1: Capacity	GDP per capita difference	Continuous	–
	Twitter followers difference	Continuous	–
	Social media index difference	Ordinal	–
H2: Shared Mission	Emergency management same	Binary	+
H3: Shared Language	Same primary language	Binary	+
H4: Geographic Proximity	Distance difference (100 miles)	Continuous	–
	Same UN region	Binary	+
	Same UN subregion	Binary	+
–	Population difference	Continuous	Control
–	Internet penetration difference	%	Control
<i>Main Effects</i>			
H1: Capacity	GDP per capita	Continuous	+
	Twitter followers	Continuous	+
–	Population	Continuous	Control

independent of the other ties in the dataset is violated. This violation occurs not only because the same actor is determining its ties to all other actors but also because structural effects such as reciprocity and transitivity can influence the propensity for ties to form. The lack of independence, therefore, prohibits the use of standard statistical models (Butts 2008; Snijders 2011).

In order to test our hypotheses, we use a statistical network model that relies on permutation tests, via a quadratic assignment procedure (QAP), to obtain estimates of statistical significance. As discussed by Lubell et al. (2012, 327), network regression with “QAP uses a bootstrapping approach randomly to ‘relabel’ the networks and examine the distribution of network statistics from the resulting population of networks.” Intuitively, in the bivariate case, the process works by permuting the rows and columns of the dependent variable. When the network is permuted, the structure of the network remains the same; only the actors playing each network position change. Thus, the dependence between the dependent variable and the independent variable is broken. By repeating the permutation procedure many times, a distribution of possible relationships between the two variables can be realized. The significance of the actual, observed relationship is then compared to the distribution of statistics produced from these randomly relabeled networks. The proportion of times a network statistic in the distribution was as extreme as the observed statistic provides an estimate of significance.

Network regression models permit the use of multiple independent variables to predict the structure of the observed network. In multivariate models, a permutation method known as double semi-partialling is used (Dekker, Krackhardt, and Snijders 2007). Given the binary outcome (i.e., follow or not follow), we use a network logistic regression model, commonly referred to as LRQAP. Lee et al. (2012) use a similar

approach in examining the antecedents of regional economic development networks in the United States. Our model was implemented in the R statistical environment (R Core Team 2013) using the *statnet* package (Handcock et al. 2008). Significance was estimated based on a distribution from 1000 permutations.

Because the dependent variable is a network structured as a matrix of follower relations, the independent variables are also in matrix form. In general, two different types of variables are used in network regression—main effects and homophily effects. Main effects measure the differences in the odds of following another actor based on the characteristics of the actors themselves. Given that a directional dyad contains the sender of a tie and the receiver of a tie, the characteristics of both organizations may matter in determining whether a tie exists. The characteristics of the sender that influence tie formation are referred to as sender effects, and the characteristics of the receiver that influence tie formation are referred to as receiver effects. For example, in the country network, countries with a higher GDP may be more likely to be followed by other actors. This receiver effect can be modeled by creating a square matrix of the same size as the dependent network that contains the GDP of each actor in the column for that given actor. A positive receiver effect would then suggest that actors with higher GDPs tend to be followed more (i.e., receive more incoming ties) than other actors. Sender effects are established in the same way, except that the attribute of interest is listed in the row of the respective actor as opposed to the column.

Beyond main effects, homophily effects may also influence tie formation tendencies. Homophily effects capture differences in the probability of tie formation that result due to actor similarity. For example, we hypothesize that actors from countries with lower GDP will be more likely to follow agencies with higher GDP. To capture this effect, a square matrix is created that indicates the difference in GDP between each actor. Thus, the i,j cell in this homophily matrix would indicate the difference in GDP between actor i and actor j . Negative cell values indicate that i has a smaller GDP than j and positive cell values indicate that i has a larger GDP than j . Therefore, we expect the coefficient on the GDP homophily term to be negative.

Given the potential correlation among a few of our predictor variables, especially with regard to distance, UN region, and UN subregion, we assessed the data for any issues of collinearity. None of the correlations among the variables was larger than 0.6.

RESULTS

To understand better network structure and composition, we first illustrated the two networks: (a) government agencies tasked as their country's lead for disaster response and recovery operations; and (b) Red Cross and Red Crescent national societies. We found profound differences between the two networks in the form of a sparse government network with relatively few connections compared with a denser (i.e., more interconnected) Red Cross and Red Crescent network. Figures 1 and 2 model the geographic relationships created by organizations following others on Twitter.³

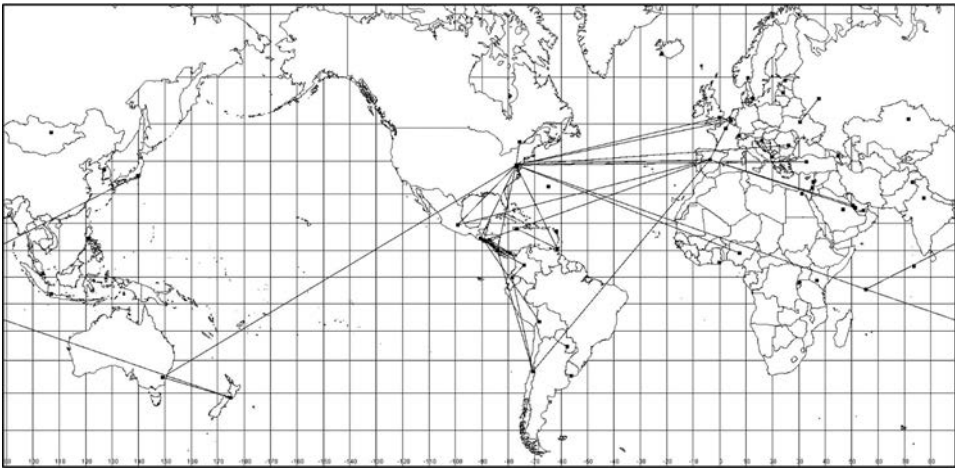


Figure 1. Twitter follower network among government agencies. Selected network cohesion measures include: Density 0.016; Average Degree 0.967.

The government agency network was relatively small; there were 60 agencies with Twitter accounts out of the relevant 185 agencies (32.4%). It was also sparse in terms of existing relationships. Of all potential ties between agencies that could exist, only 1.6% actually occurred. Furthermore, the average degree centrality is less than one (0.967), which indicates that several agencies (20) neither followed nor were followed by any other agency. Those isolates (i.e., disconnected agencies) therefore failed to contribute to the larger knowledge network and, in fact, did not actually join.

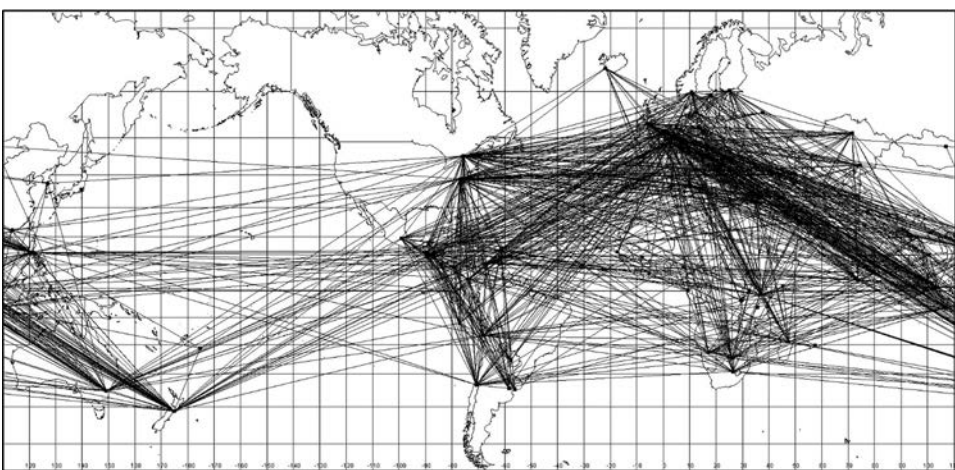


Figure 2. Twitter follower network among Red Cross and Red Crescent national societies. Selected network cohesion measures: Density 0.150; Avg Degree 14.402.

Notable examples include India, Japan, Kenya, Norway, and Pakistan. Conversely, Table 3 reports both the agencies most followed and the agencies that most actively sought out others to follow.

In comparison to the government agency network, the Red Cross and Red Crescent knowledge network is larger (97 organizations) and denser. Of all of the potential ties between agencies that could exist, 15.0% actually occurred, a much higher percentage than in the government agency network. Furthermore, only six failed to follow or be followed by another agency. On average, national societies followed 14.4 others, which created a much more interconnected network. The result is a more robust transnational knowledge network in which message content can be monitored more extensively and on a global basis. Table 4 reports both the agencies most followed and the agencies that actively sought out others to follow.

One factor that may explain the difference between the Red Cross and Red Crescent network and the government agency network is the role played by the IFRC in advocating for social media use. The IFRC and other innovative members have invested considerable time and resources to promote social media use across the national societies (American Red Cross 2012; IFRC 2009). There does not appear to be a comparable advocate for social media in the government network.

Despite the structural differences in the two networks, we found commonalities with respect to antecedents. In other words, despite differences in the overall tendencies for ties to form in government network and Red Cross and Red Crescent network, the organizing factors were quite similar. Shared factors include geographic proximity and common language. In addition, the number of followers that an organization had also proved to be a statistically significant determinant of an organization’s propensity to send and receive ties. Table 5 displays the model outputs for both networks. For each model, the first column indicates the coefficient and significance level for the independent variables. The second column provides the odds ratio and thus offers a more intuitive understanding of the size of the effect.

In both networks, the closer two organizations were in terms of physical distance, the more likely they were to follow one another. This finding clearly supports our fourth hypothesis. In the government agency network, an example is the propensity of several Central American countries to monitor each other’s accounts (see Figure 1), likely due to their common exposure to hurricanes and flooding, which led to a collective sense of shared risk.

TABLE 3
Key Organizations in Government Agency Network

<i>Country</i>	<i>Followed By</i>	<i>Country</i>	<i>Following</i>
United States	13	Costa Rica	8
Guatemala	5	Spain	7
El Salvador	5	Guatemala	6
Costa Rica	4	El Salvador	6
Honduras	4	Panama	4

TABLE 4
Key Organizations in Red Cross and Red Crescent Network

<i>Country</i>	<i>Followed By</i>	<i>Country</i>	<i>Following</i>
United States	67	France	59
Canada	60	Canada	51
Britain	59	Australia	47
France	56	Kenya	41
Philippines	55	New Zealand	41

TABLE 5
QAP Logistic Regression Results

	<i>Govt Agency Network</i>		<i>Red Cross Network</i>	
	<i>Estimates</i>	<i>Odds ratio</i>	<i>Estimates</i>	<i>Odds ratio</i>
(Intercept)	−11.7970 *		−9.4435 ***	
<i>Homophily Effects</i>				
GDP per capita	−0.0030	0.997	−0.0015 *	0.998
Twitter followers	−0.0002	0.999	−0.0001	0.999
Social media index	0.0550	1.057	0.0242	1.024
Emergency mgmt same	0.8927 *	2.442	N/A	N/A
Same primary language	2.1272 **	8.391	1.3866 ***	4.001
Distance	−0.0403 ***	0.961	−0.0159 ***	0.984
Same UN region	−0.8085	0.446	−0.3176	0.728
Same UN subregion	1.5064	4.510	0.4671 *	1.595
Population	−0.0292	0.971	−0.0003	0.999
Internet penetration	−0.0121	0.988	0.0019	1.002
<i>Main Effects</i>				
Receiver: GDP per capita	−0.7852	0.456	−0.0161	0.0984
Sender: GDP per capita	0.7715	2.163	0.300	1.350
Receiver: Twitter followers	0.4317 ***	1.540	0.4130 ***	1.511
Sender: Twitter followers	0.2901 *	1.337	0.3413 ***	1.407
Receiver: Population	0.0892	1.093	0.0069	1.007
Sender: Population	0.0844	1.088	−0.152	0.859
AIC	361.5841		5711.574	
BIC	466.5061		5825.799	
Deviance	327.5841		5679.574	
Null deviance	4907.482		12909.1731	
Num. obs.	3540		9312	

Note: For the main effects of GDP per capita, Twitter followers, and population (100,000 s), the values were log transformed. For the homophily effects, which rely on difference scores for those variables, the values were scaled by dividing by 100.

*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Red Cross and Red Crescent organizations also demonstrated a tendency to interact based on geographic proximity. In addition, for Red Cross and Red Crescent organizations, not only was geographic distance statistically significant, but so too was whether organizations were located within the same UN subregion. Neither network was bound by geography alone, however. In both networks, popular organizations emerged that garnered attention from around the world. The Federal Emergency Management Agency (FEMA) in the United States, for example, elicited the most attention in the government agency network (13 followers). And in the Red Cross and Red Crescent network, the US (67 followers), Canada (60 followers), the UK (59 followers), and France (56 followers) were the most popular countries to follow.

In addition to geographic proximity, common language helps to account for the variation in our model. In both networks, organizations were more likely to follow others who share their primary language, which supports our third hypothesis. A visual inspection of the government agency network offers further evidence. Many Latin American countries, for example, followed each other as well as Spain, and several English-speaking countries (e.g., Australia, Canada, New Zealand, Nigeria, and the United Kingdom) followed the United States. Red Cross and Red Crescent national societies demonstrated a similar tendency to rely on common language. While this tendency to interact around a shared language may not be surprising, the size of the effect is worth noting. For the government agency network, the odds of following another agency are 8.3 times higher for dyads with a common language compared to those without. For the Red Cross network, common language increased the odds of following by 4.0 times. Such large effect sizes are surprising given that Twitter has developed a tweet translation function that currently operates for 40 different language pairs.

In addition to the homophily effects of geographic proximity and common language, one main effect, the number of followers an organization possessed, proved to be statistically significant in both models. Organizations with more overall followers (e.g., citizens, news outlets, and other organizations) were likely to both follow more accounts and receive more followers from within their respective transnational knowledge network. Follower counts may indicate the value people placed on accounts, content, as well as the reach that agency achieved. If follower counts are proxies for organizational capacity and/or information quality, then this result supports our first hypothesis. Additionally, in the Red Cross and Red Crescent network, per-capita GDP was statistically significant. Because of the resources available in these wealthier countries, we assume that these agencies generated valuable content that other agencies wanted to consume. Developing countries may have also followed wealthier countries with the hope of garnering resources and support in the event of a disaster.

Finally, due to the variability in missions assigned to government agencies (e.g., emergency management, domestic security, public health, public safety, and environmental protection), we included the variable shared mission in the government agency network model. We found that government agencies that focused solely on emergency management were more likely to follow other agencies with a sole focus

on emergency management compared to agencies responsible for a more diverse set of tasks. This supports our second hypothesis, in which we surmised that agencies' seek out relevant information and try to avoid unrelated message content that may clutter their Twitter feed. Because Red Cross and Red Crescent national societies were more homogenous, we did not attempt to incorporate the variable in that model.

DISCUSSION AND CONCLUSION

Our findings support several conclusions from related research on not only transnational knowledge networks specifically, but also collaborative governance, TNGOs, and international political economy more generally. Most importantly, we contribute to the understanding of how knowledge networks form globally and how social media can enable that process. Much of the research on both nation-based (Dawes 2009; Gil-Garcia et al. 2010; Gil-Garcia 2012) and transnational (Dawes et al. 2012; Navarrete 2009) knowledge networks rely on case study methodology to examine antecedents. Our study uses LRQAP analysis to test the generalizability of some of those propositions, while also accounting for additional micro-level processes occurring in our networks. What our findings suggest is that shared mission helps to bridge the contextual distances reported by Dawes et al. (2012). Our findings also indicate that organizations are more likely to follow others demonstrating high levels of capacity that perhaps produce higher-quality information.

Despite the potential of social media to facilitate information exchange regardless of physical distance (see Mergel 2012; Roberts 2011), organizations were still more likely to follow neighbors and others in close geographic proximity. It is logical in this policy domain to assume that neighbors would share common problems (e.g., exposure to shared risk) and, therefore, would be more likely to interact (Comfort 1999). This finding conforms to a long history of research that demonstrates that two actors operating in close proximity are more likely to interact. Examples in the interorganizational context include international trade (Gowa 1994) and inter-local cooperation (Lee et al. 2012).

In line with previous research (e.g., Navarrete et al. 2010; Roth and Sever 2007), shared language positively affected whether organizations sought out each other's messages. The tendency to connect with others who share the same language and operate in close geographic proximity consequently creates regional clusters of organizations that may constrain the network's ability to widely disseminate critical pieces of information. Interestingly, social media platforms hold promise in further breaking down these barriers. Both Twitter and Facebook messages have options for translation into other languages. However, while language is less of a technical barrier for information retrieval than in the past, it nonetheless shapes the network formation in both cases examined in this article.

Some social media accounts proved to be more popular than others and transcended language and geographic barriers. FEMA, for example, was the most followed account in the government agency network. In the Red Cross and Red Crescent

network, the US, Canada, the UK, and France were popular. All of these organizations exhibited indicators conducive with high-capacity organizations (e.g., wealthy countries, large follower counts, and high social media index scores). Our model demonstrated that organizations were generally more likely to follow high-capacity organizations, which is consistent with offline behaviors in which organizations seek out actors thought to possess needed information, personnel, and resources (Comfort 1999; Kapucu 2006; 2009).

The two networks demonstrated notable structural differences. The higher participation rate by Red Cross and Red Crescent national societies suggests a more robust, interactive knowledge network in which information should be expected to circulate more evenly. In the government agency network, the lack of participation and connectivity may deny organizations from obtaining critical pieces of information, conditions that lead to suboptimal disaster outcomes (Comfort et al. 2010; Kapucu 2006; Waugh 2009).

Because of our case selection, we were able to account for the role of external advocates in promoting the formation of knowledge networks. The IFRC promoted social media use as a core competence for national societies, whereas the sparser government agency network was more self-organized and lacked a central champion for change and social media use. Future research would benefit from explicitly operationalizing this concept, perhaps through measures of participant perception. Furthermore, the differences between our two networks were likely due to more than the presence of the IFRC and other influential organizations. Additional differences likely lie in the missions of the organizations themselves. Government agencies, for example, focus their attention on their countries and are accountable solely to domestic institutions and officials; therefore, they concern themselves less with international incidents than the Red Cross and Red Crescent—organizations imbued with a sense of global humanitarian action (IFRC 2009). Those nongovernmental organizations also seek to develop their social media capabilities in order to raise funds, recruit volunteers, and increase the public's awareness of their mission (see Briones et al. 2011; Wukich and Khemka 2016). Government agencies often do not operate under the same resource development pressures.

A notable limitation of our study is our failure to explicitly operationalize information quality. We rely on measures of capacity (e.g., followers) as a proxy. Future studies could differentiate types of messages and assess their quality based on established best practices in emergency communications (Lindell and Perry 2012; Sutton et al. 2014). Researchers could then empirically assess whether the producers of such high-quality information are rewarded with more followers.

We acknowledge that high-quality information might not necessarily be driving others to seek out these organizations. Another reason for their popularity may be a more explicit manifestation of resource dependence (Pfeffer and Salancik 1978). Organizations may seek out wealthier partners to develop other types of relationships, not just for the information they provide on social media. This provides an opportunity for future research. Indeed, information sharing can lead to other types of cooperation. Here, Isett and Provan's (2005) notion of multiplexity is germane, where multiple types of cooperation can occur simultaneously and cooperation in

one area can lead to cooperation in another. Kapucu and Hu (2016) applied the concept to emergency management, and future research could elaborate on their findings by including social media as a type of cooperative activity. This point can be expanded to capture the role of existing offline/face-to-face interactions and how they might lead to online relationships.

Another limitation of this article is our failure to operationalize each contextual factor that Dawes et al. (2012) proposed in their model of transnational network participation. While we addressed key factors such as mission, language, geographic proximity, and the availability of resources, we do not explicitly account for factors such as organizational culture, trust and past relationships, executive support and organizational commitment, and organizational rules and procedures. Such factors comprise what Bardach (1998) referred to as collaborative capacity and deserve attention in this context. The role of public managers in this process should not be ignored because they can facilitate information exchange by (a) first supporting the adoption and implementation of social media within their organization; (b) creating policy and procedures that empower personnel to seek out information from a variety of sources; (c) allocating the requisite personnel to implement that policy; and (d) incorporating social media data into organizational decision-making processes. This process may require time and resources, and can include trial-and-error experimentation in order to determine how to integrate social media into existing operations in the most appropriate manner (Mergel and Bretschneider 2013). Future research could explore those determinants using survey methodology and/or multiple-case case study approaches.

Future research could also test whether our results are generalizable to other policy domains. In emergency management, the operational impetus for cooperation and information exchange is relatively clear—reduce shared risk and protect lives and property. Organizations generally recognize the importance of at least information sharing in the interorganizational context. This phenomenon may also be the case for public health and domestic security, but perhaps not for organizations whose missions relate to other policy domains such as transportation, economic development, and education. Research is needed in these areas.

Finally, this article illuminates the potential for expanded transnational knowledge sharing via social media platforms. Research has focused on a number of impediments to information-sharing projects, several related to the transaction costs of aligning goals, negotiating terms, and acquiring interoperable technical infrastructures (see Gil-Garcia 2012; Gil-Garcia et al. 2007; 2010; Navarrete et al. 2010;). Social media platforms considerably reduce those costs because they provide a relatively inexpensive entry into a knowledge network. Social media may not be the appropriate solution for every information sharing need, especially when dealing with sensitive information; however, they do provide increasingly popular and accessible options.

NOTES

1. This type of information technology is commonly referred to as information and communication technology throughout Europe.

2. Sources included the UNISDR (<http://www.unisdr.org/partners/countries>), the European Commission's Humanitarian Aid and Civil Protection (http://ec.europa.eu/echo/files/civil_protection/vademecum/fr/2-fr-4.html#list), Asian Disaster Reduction Center (<http://www.adrc.asia/disaster/index.php>), Caribbean Disaster Emergency Management Agency (<http://www.cdema.org>), and PreventionWeb (<http://www.preventionweb.net>).

3. We visually modeled the two networks based on geographic location using the software program OraGIS (Carley et al. 2013). We identified network cohesion statistics (e.g., density and average degree centrality) using UCINET (Borgatti, Everett, and Freeman 2002).

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APPENDIX I

Government Agency Accounts

<i>Country</i>	<i>Organization</i>	<i>Twitter Account</i>
Albania	Albanian Civil Protection (Mbrojtja Civile Shqiptare)	MbrojtjaCivile
Antigua and Barbuda	National Office of Disaster Services (NODS)	NODS7
Australia	Australian Emergency Management	AEMKH
Bahrain	Ministry of Interior	moi_bahrain
Belgium	Crisiscentrum/Centre de Crise	CrisiscenterBE
Bermuda	Emergency Measures Organisation	EMOBermuda
Bolivia	Viceministerio de Defensa Civil	DefensaCivilBol
Canada	Public Safety Canada	Safety_Canada
Chile	Oficina Nacional de Emergencia (ONEMI)	onemichile
Colombia	Dirección de Gestión del Riesgo (DGR)	ungrd
Costa Rica	Comisión Nacional de Prevención de Riesgos y Atención de Emergencias (CNE)	CNECostaRica
Croatia	National Protection and Rescue Directorate (NPRD)	DUZS_HR
Denmark	Danish Emergency Management Agency	DEMA_eng
Ecuador	Secretaría Nacional de Gestión de Riesgos (SNGR)	RiesgosEc
Egypt	Cabinet Information and Decision Support Center, Crisis Management and DRR Sector (IDSC)	IDSC_Egypt
El Salvador	Dirección General de Protección Civil, Prevención y Mitigación de Desastres	PROCIVILSV
Estonia	Estonian Rescue Board (Päästeamet)	paasteamet
France	Directorate General Civil Protection and Crisis Management (Direction Générale de la Sécurité Civile et de la Gestion des Crises)	Place_Beauvau

(Continued)

<i>Country</i>	<i>Organization</i>	<i>Twitter Account</i>
Georgia	Ministry of Environment and Natural Resources Protection of Georgia (MOE)	MOEgovge
Ghana	National Disaster Management Organisation (NADMO)	NADMOGHANA
Guatemala	Coordinadora Nacional para la Reducción de Desastres (CONRED)	ConredGuatemala
Honduras	Comisión Permanente de Contingencias (COPECO)	COPECO_HONDURAS
Iceland	Department of Civil Protection and Emergency Management	almannavarnir
India	National Disaster Management Authority (NDMA)	NDMA_INDIA
Indonesia	National Disaster Management Agency (BNPB: Badan Nasional Penanggulangan Bencana)	BNPB_Indonesia
Jamaica	Office of Disaster Preparedness and Emergency Management	odpem
Japan	Fire and Disaster Management Agency	FDMA_JAPAN
Kazakhstan	Ministry of Emergency Situations (EMER)	emer_kz
Kenya	National Disaster Operation Centre	NDOCKenya
Latvia	State Fire and Rescue Service of Latvia	ugunsdzeseji
Lebanon	Civil Defence	CivilDefenseLB
Maldives	National Disaster Management Center	NDMC_Official
Mexico	Sistema Nacional de Protección Civil (SINAPROC)	PcSegob
Mongolia	National Emergency Management Agency (NEMA)	amynra777
New Zealand	Ministry of Civil Defence and Emergency Management	Nzcivildefence
Nicaragua	Sistema Nacional para la Prevención, Mitigación y Atención de Desastres (SINAPRED)	SINAPRED
Nigeria	National Emergency Management Agency (NEMA)	nemanigeria
Norway	Directorate for Civil Protection and Emergency Planning, Ministry of Foreign Affairs (DSB)	DSB_no
Pakistan	National Disaster Management Authority (NDMA)	PakistanNDMA
Panama	Sistema Nacional de Protección Civil (SINAPROC)	SINAPROC_PANAMA
Paraguay	Secretaria de Emergencia Nacional (SEN)	senparaguay
Philippines	National Disaster Risk Reduction and Management Council (NDRRMC)	NDRRMC_OpCen
Qatar	Ministry of Interior: Civil Defence	MOI_QatarEn
Republic of Korea	National Emergency Management Agency (NEMA)	Nema_SafeKorea
Romania	General Inspectorate for Emergency Situations (IGSU)	IGSU_URGENTE_RO

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<i>Country</i>	<i>Organization</i>	<i>Twitter Account</i>
Russian Federation	Ministry of Civil Defence, Emergencies and Disaster Relief of the Russian Federation	MchsRussia
Rwanda	Ministry of Disaster Management and Refugee Affairs	MIDIMAR_gov
Saudi Arabia	General Directorate of Civil Defense	MOIeServices
Seychelles	Department of Risk and Disaster Management	DRDM_Seychelles
Singapore	Singapore Civil Defence Force	SCDF
Spain	Dirección General de Protección Civil y Emergencias	INTERIORGOB
State of Palestine	General Directorate of Civil Defense, Ministry of the Interior	pcd102
Switzerland	Civil Protection (Das Bundesamt für Bevölkerungsschutz BABS)	vbs_ddps
Trinidad and Tobago	Office of Disaster Preparedness and Management	ODPM_TT
Turkey	Prime Ministry, Disaster and Emergency Management Presidency (AFAD)	afadbaskanlik
Ukraine	Ministry of Emergencies and Affairs of Population Protection from the Consequences of Chernobyl Cat astrophe of Ukraine	MNS-UA
United Arab Emirates	General Directorate of Civil Defense	DCDDubai
United Kingdom	Civil Contingencies Secretariat (CCS)	UKResilience
United States of America	Federal Emergency Management Agency	fema
Uruguay	Sistema Nacional de Emergencias (SINAE)	SINAE2013

APPENDIX II

Red Cross and Red Crescent Accounts

<i>Country</i>	<i>Organization</i>	<i>Twitter Account</i>
Algeria	Croissant Rouge DZ	CRA_DZ
Andorra	Andorran Red Cross	creuroja
Argentina	Cruz Roja Argentina	CruzRojaArg
Armenia	Armenian Red Cross Society	RedCrossArmenia
Australia	Australian Red Cross	RedCrossAU
Austria	Rotes Kreuz	roteskreuzat
Azerbaijan	Azerbaijan Red Crescent Society	AzRedCrescent
Bahamas	Bahamas Red Cross	BahamasRedCross
Bahrain	Bahrain Red Crescent Society	RCSBahrain

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<i>Country</i>	<i>Organization</i>	<i>Twitter Account</i>
Barbados	Barbados Red Cross Society	BdosRedCross
Belgium	Croix Rouge	CroixRougeBE
Belize	Belize Red Cross	BelizeRedCross
Bolivia	Cruz Roja Bolivia	cruzrojabolivia
Botswana	Botswana Red Cross Society	BotswanaRCS
Bulgaria	Bulgarian Red Cross Society	BGRedCross
Burundi	Croix Rouge Burundi	CroixRougeBDI
Canada	Canadian Red Cross	redcrosscanada
Chile	Cruz Roja Chilena	CruzRojaInforma
Macao (Special Administrative Region)	Macau Red Cross	MacauRedCross
Colombia	Cruz Roja Colombia	cruzrojacol
Comoros	Croissant-Rouge Comorien	CRComorien
Costa Rica	Cruz Roja Costarricense	CruzRojaCRC
Cote d'Ivoire	Croix-Rouge Côte d'Ivoire	CroixRougeCI
Denmark	Danish Red Cross	AndersLadekarl
Dominica	Cruz Roja Dominicana	crdominicana
Ecuador	Cruz Roja Ecuador	cruzrojaecuador
El Salvador	Cruz Roja El Salvador	CruzRojaSal
Finland	Finish Red Cross	PunainenRisti
France	Croix-Rouge Française	CroixRouge
Gabon	Croix-Rouge Gabonaise	Croix_Rouge_Gab
Georgia	Red Cross Society of Georgia	RedCrossGeorgia
Germany	Deutsches Rotes Kreuz	roteskreuz_de
Guatemala	Cruz Roja Guatemala	CRGuatemalteca
Haiti	Croix-Rouge Haitienne	KwaWouj
Honduras	Cruz Roja Hondureña	cruzrojahon
Iceland	Icelandic Red Cross	raudikrossinn
India	Indian Red Cross Society	IndianRedCross
Indonesia	Indonesian Red Cross Society	palangmerah
Iraq	Iraqi Red Crescent Society	iraqircs
Ireland	Irish Red Cross	irishredcross
Israel	Magen David Adom	MDAVolunteers
Italy	Croce Rossa Italiana	crocerossa
Jamaica	Jamaica Red Cross	Jamaicaredcross
Japan	Japan Red Cross	JRCS_PR
Kazakhstan	Kazakh Red Crescent	KazakhRCS
Kenya	Kenya Red Cross	KenyaRedCross
Kuwait	Kuwait Red Crescent Society (KRCS)	kracs_org_kw
Kyrgyzstan	Red Crescent Society of Kyrgyzstan	redcrescentkg
Lebanon	Lebanese Red Cross	LRCEMS
Luxembourg	Croix-Rouge Luxembourgeoise	CroixRougeLu
Madagascar	Malagasy Red Cross	MadaRedCross
Malaysia	Malaysia Red Crescent Society	MRC_Penang
Maldives	Maldivian Red Crescent	maldivianrc

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<i>Country</i>	<i>Organization</i>	<i>Twitter Account</i>
Mali	Croix-Rouge Malienne	CroixRougeMali
Malta	Red Cross Malta	Maltaredcross
Mauritius	Mauritius Red Cross Society	MRURedCross
Mexico	Cruz Roja Mexicana	CruzRoja_MX
Mongolia	Mongolian Red Cross Society	RedCrossmn
Myanmar	Myanmar Red Cross	MyanmarRedCross
Namibia	Red Cross Society of Namibia	NamibiaRedCross
Nepal	Nepal Red Cross Society	NepalRedCross
Netherlands	Rode Kruis	RodeKruis
New Zealand	New Zealand Red Cross	NZRedcross
Nigeria	Nigerian Red Cross Society	ncrs_ng
Norway	Røde Kors Norge	rodekorsnorge
Pakistan	Pakistan Red Crescent Society	PRCS_official
Panama	Cruz Roja Panameña	CruzRojaPanama
Paraguay	Cruz Roja Paraguaya	CruzRojaPy
Peru	Cruz Roja Peruana	CruzRojaPeru
Philippines	Philippine Red Cross	philredcross
Portugal	Cruz Vermelha Portuguesa	cvpsedenacional
Qatar	Qatar Red Crescent	QRCS
Republic of Korea	South Korea Red Cross	KoreanRedCross
Romania	Crucea Rosie Romana	CruceaRosie
Russian Federation	Russian Red Cross	MKKK
Senegal	Croix-Rouge Senegal	crssenegal
Singapore	Singapore Red Cross	SGRedCross
Somalia	Ururka Bisha Cas (SRCS)	Bishacas1
South Africa	South African Red Cross Society	SARedCross
Spain	Cruz Roja Española	CruzRojaEsp
Sri Lanka	Sri Lanka Red Cross	SLRedCross
State of Palestine	Palestine Red Crescent Society	PalestineRCS
Sweden	Svenska Röda Korset	RodaKorset
Switzerland	Swiss Red Cross	redcross_ch
Syrian Arab Republic	Syrian Arab Red Crescent	SYRedCrescent
	Organization	
Thailand	Thai Red Cross	ThaiRedCross
The Former Yugoslav Republic of Macedonia	The Red Cross of the Former Yugoslav Republic of Macedonia	crvenkrstrm
Tonga	Tonga Red Cross	Redcross4
Tunisia	Croissant Rouge Tunisien	CroissantRouge
Turkey	Turkish Red Crescent	TrRedCrescent
Uganda	Uganda Red Cross Society	UgandaRedCross
United Arab Emirates	Red Crescent UAEU	RedCrescentUAEU
United Kingdom	British Red Cross	BritishRedCross
Tanzania	Tanzania Red Cross Society	tcrl
United States of America	American Red Cross	RedCross
Uruguay	Cruz Roja Uruguay	cruzrojauruguay
Venezuela	Cruz Roja Venezolana	Crvoficial